



A Review: Microstructural and Phase Evolution in Alloys during Extended Plastic Deformation

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Materials in service and during processing are often subjected to plastic deformation. For multi-phase metallic alloys, simple geometric models and atomistic simulations suggest that two distinctive regimes in these materials' evolution during deformation exist. At low strains, evolutions are often dominated by the kinetic roughening of interfaces, which results from the superdiffusive transport of matter in sheared crystals. At high strains and temperatures where thermal diffusion is sluggish, on the other hand, shearing-induced forced atomic mixing dominates these evolutions, resulting in significantly enhanced solubility. Distinguishing these two regimes is shown to provide a convenient framework for rationalizing and analyzing recent experiments and simulations on wear of layered structures in the low strain-regime and nonequilibrium phase co-existence and self-organization in highly immiscible or reactive alloy systems in the high-strain regime.

INTRODUCTION

Metallic alloys are routinely subjected to extensive plastic deformation. This is the case during alloy synthesis by ball milling and mechanical alloying; during shaping processes such as rolling, extrusion, drawing and forging; during joining by roll bonding and friction stir welding. Plastic deformation is similarly important for materials in service owing to creep, fatigue or sliding and impact wear. In crystalline solids plastic deformation can directly drive microstructural evolution through the creation and motion of dislocations, for instance resulting in grain size evolution by recovery and recrystallization. In addition, plastic deformation can alter local alloy compositions and chemical order, resulting in phase transformations and phase evolutions; it is this latter behavior that is the subject of this article. We will concentrate mostly on the case where thermal diffusion plays a negligible role in these evolutions, i.e., for large values of the Zener-Hollomon parameter $Z = (\text{strain rate})/(\text{diffusion coefficient})$. The motivation for this choice is

three-fold: First, this regime allows for isolating the effects directly arising from plastic deformation, many of which will be operating regardless of the deformation temperature; second, many of the shaping and joining techniques and the degradation processes listed above are often taking place at or near room temperature (RT), and thus at low homologous temperatures in many engineering materials, where thermal diffusion of point defects is negligible. Third, plastic deformation and severe plastic deformation (SPD) at such low homologous temperatures provide a powerful route for fabricating nanostructured metals and alloys with novel and attractive properties. The scope of this topic is quite broad, and the interested reader is directed to the many books published on this subject in the last 3 decades^{1,2} as well as those dedicated to specific topics such as high-pressure torsion,^{3–5} mechanical alloying^{6,7} and wear.^{8,9} The emphasis in the present article will be placed on the fundamental mechanisms underlying phase and interface evolutions in two-phase alloys subjected to plastic deformation. In “Stages of Evolution in Multi-Phase Alloys Subjected to Plastic Deformation” section we will use simple models to establish the presence of two

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distinctive regimes, a first one at low strain that is related to the morphological evolution and instability of interfaces and a second regime at large strain that is concerned with enhanced solubility limits due to forced atomic mixing. These two regimes will then be discussed in “[Low-Strain Regime: Morphological Instabilities](#)” and “[High-Strain Regime: Forced Atomic Mixing and Self-Organization](#)” sections, respectively, drawing from recent experiments, modeling and atomic simulations, before offering some final perspectives in “[Conclusion](#)” section.

STAGES OF EVOLUTION IN MULTI-PHASE ALLOYS SUBJECTED TO PLASTIC DEFORMATION

A simple rigid lattice model reveals that the evolution of two-phase crystalline alloys subjected to plastic deformation at low temperature proceeds in two distinctive stages as plastic strain increases.^{10,11} Consider, for example, an initially flat chemical interface separating elements A and B (Fig. 1a): The shearing of that interface by a dislocation of Burgers vector $\mathbf{b}$ produces a step $\mathbf{b}$ on the interface as long as $\mathbf{b}$ is not contained in the interface plane, which thus requires that the glide plane is not parallel to the initial interface. At low strain, the evolution of the interface is controlled by this out-of-plane shearing, and it results in the roughening of the interface; see Fig. 1b,c. The kinetic roughening is characterized by undulations whose dominant wavelength is approximately the glide distance of dislocations, i.e., the system size in Fig. 1, since periodic boundary conditions were employed in this simulation. In polycrystalline materials, the dominant roughening wavelength is the grain size, and that length scale can exceed microns. While out-of-plane shearing dominates the early part of kinetic roughening, in-plane shearing (if available) contributes to the roughening as well, in particular to the formation of overhangs. Similar wavy patterns are observed in geological systems probably for related reasons, albeit at much larger scales and by different deformation

mechanisms.^{12–15} We will come back to this analogy in “[Low-Strain Regime: Morphological Instabilities](#)” section. At larger strains the system enters a second regime: Shearing involving distinct slip systems produces atomic mixing, see Fig. 1d, and eventually it leads to the homogenization of the alloy configuration at steady-state, Fig. 1f. Notice that the rate of this forced atomic mixing, for example characterized by the number of A-B bonds, initially increases super-linearly with strain since mixing is proportional to the interface area, which is increasing with strain due to kinetic roughening. The transition from kinetic roughening to atomic mixing, which takes place for strain values of ≈ 10 –100 in the example illustrated by Fig. 1, will depend on the initial density of interfaces; the larger the initial domain size, the larger the strain required to randomize a chemical structure. For example, the characteristic plastic strain required to dissolve a monodisperse distribution of spherical particles of radius R_p in the absence of thermal diffusion scales linearly with R_p , i.e.,

$$\Delta\varepsilon = \frac{16c_B(1-c_B)}{3} \frac{R_p}{V_p} \frac{R_p}{b} \quad (1)$$

where c_B is the concentration of solute B, V_p is the volume fraction of precipitates, and b is the modulus of the Burgers vector.¹⁶ For a diffusive process, in contrast, the strain required for homogenization scales with R_p^2 .

Diffusive mixing, in fact, would lead to a very different interfacial evolution. In that case, kinetic roughening would be absent or minimal, and the interface would directly evolve from flat and sharp to flat and diffuse, with the diffuseness (again measured by the number of A-B bonds, initially increasing as $(\text{time})^{1/2}$; see Fig. 2. The essential feature of plastic deformation responsible for this difference is that each shear introduces a macroscopic perturbation to the alloy configuration. For that reason, transport and mixing take place by convective flow. Richardson identified that in such systems mixing is superdiffusive, i.e., the relative

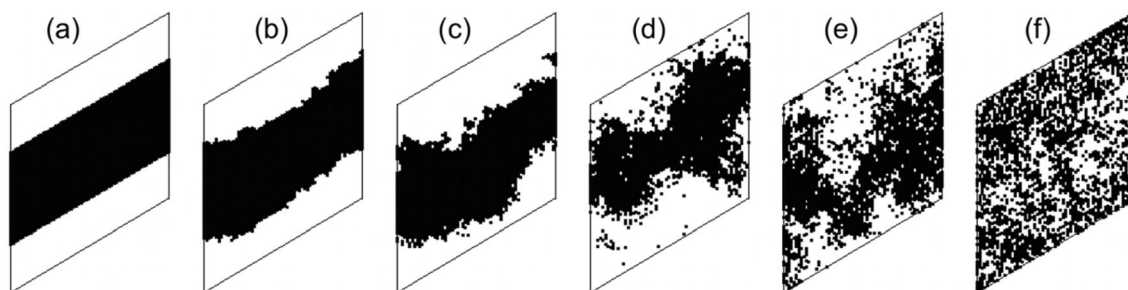


Fig. 1. Monte Carlo simulations of the evolution of an initially perfectly flat (111) interface in a face-centered cubic lattice subjected to isotropic shearing at a temperature where thermal diffusion is negligible. The initial layers are each comprised of 32 (111) planes, thus corresponding to a thickness of ≈ 6.6 nm in Cu. Snapshots of one (111) plane at increasing shear strains of (a) 0, (b) 1.36, (c) 5.78, (d) 21.8, (e) 39.1, and (f) 90.6. Adapted from Ref. 11.

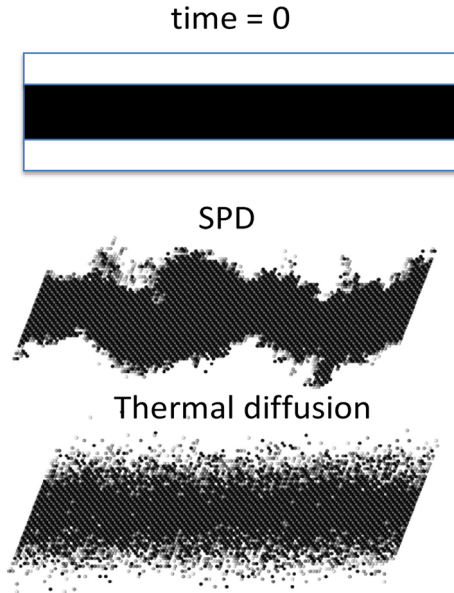


Fig. 2. Monte Carlo simulations of the evolution of an initially perfectly flat interface (top) subjected to severe plastic deformation (center) or thermal annealing (bottom). Reprinted with permission from Ref. 18.

mean square displacement between markers (or Richardson pairs), in our case atoms, grows with time as $(\text{time})^n$ where the exponent $n > 1$.¹⁷ The rigid-lattice shearing model introduced in Fig. 1 yields $n = 2$.

Superdiffusivity is also reflected in the fact that two markers move away from each other at a rate that increases with their initial separation distance since the larger that distance, the more likely it is that dislocations will pass between them. Superdiffusivity can be quantified by an effective pair diffusion coefficient defined as the relative mean square displacement between two markers during fixed (and usually small) time or strain increments. A direct analysis of plastic deformation in the above lattice model indicates that this effective diffusion coefficient is expected to increase linearly with the pair separation distance r_{ij} and the plastic strain rate: $D_{\text{eff}}^{\text{shear}} = |b| r_{ij} \dot{\gamma}_{\text{pl}}$.¹⁹ This linear dependence is indeed observed in KMC and MD simulations,²⁰ as illustrated in Fig. 3. When the pair separation distance is so large that the two markers move independently from one another, for instance, when r_{ij} exceeds the grain size, a cross-over takes place from the superdiffusive to a diffusive regime and now $D_{\text{eff}}^{\text{shear}} \approx 2D^*$, where D^* is the tracer diffusion coefficient.¹⁶ This analysis can be extended to the case of amorphous materials. In these systems plastic deformation at the atomic scale takes place by the formation of shear transformation zones (STZs), whose extensions are ≈ 1 nm, for instance, see Refs. 21,22. As a result, the cross-over from superdiffusive to diffusive mixing takes place at a similar and thus very short length scale.¹⁶ Kinetic

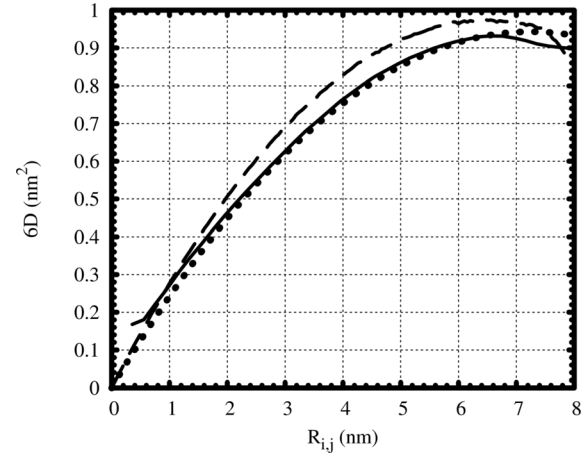


Fig. 3. Effective pair diffusion coefficient as a function of r_{ij} , the pair separation distance between two markers i and j . Comparison among KMC simulations (dashed line), MD simulations (solid line), and the expression expected for a system with periodic boundary conditions (dotted line). Reprinted with permission from Ref. 20.

roughening should therefore be quite limited for amorphous solids subjected to plastic deformation. MD simulations by Fu, Falk and Rigney are indeed consistent with this expectation; see Fig. 4.²³

We note that if plastic deformation takes place at a temperature where thermal diffusion is no longer negligible, the above immiscible A-B alloys under “geometric” shearing would reach a self-organized steady state comprised of finite scale co-existence between A-rich and B-rich domains, See Fig. 5a. This self-organization reaction into compositional patterns is a direct consequence of the kinetic roughening of interfaces and its competition with thermal diffusion:^{10,24} As illustrated in Fig. 5b, the outcome of this competition between atomic mixing forced by plastic deformation and decomposition promoted by thermal diffusion depends on the length scale of interest. The length scale where diffusion from these two contributions crosses over defines a stable composition domain size, i.e., the characteristic length scale of the self-organized patterns. This self-organization may be relevant to various high-temperature processes such as shaping and joining in the solid state, e.g., friction stir welding. Self-organization can even take place in the absence of thermal diffusion, but for different reasons, as will be discussed in “[High-Strain Regime: Forced Atomic Mixing and Self-Organization](#)” section.

Effective pair-diffusion coefficients provide direct insight into the mechanisms of plastic deformation, in particular whether deformation is homogeneous or not. For instance, MD simulations of a spherical inclusion of species $X = \text{Cu, Ag, Ni}$ in a Cu matrix subjected to cyclic biaxial compression reveal key differences between $D_{\text{eff}}^{\text{shear}}$ for Cu-Cu, Cu-X and X-X atom-pairs of nearest neighbor markers.²⁰ For the Cu-Cu system, deformation is homogeneous as

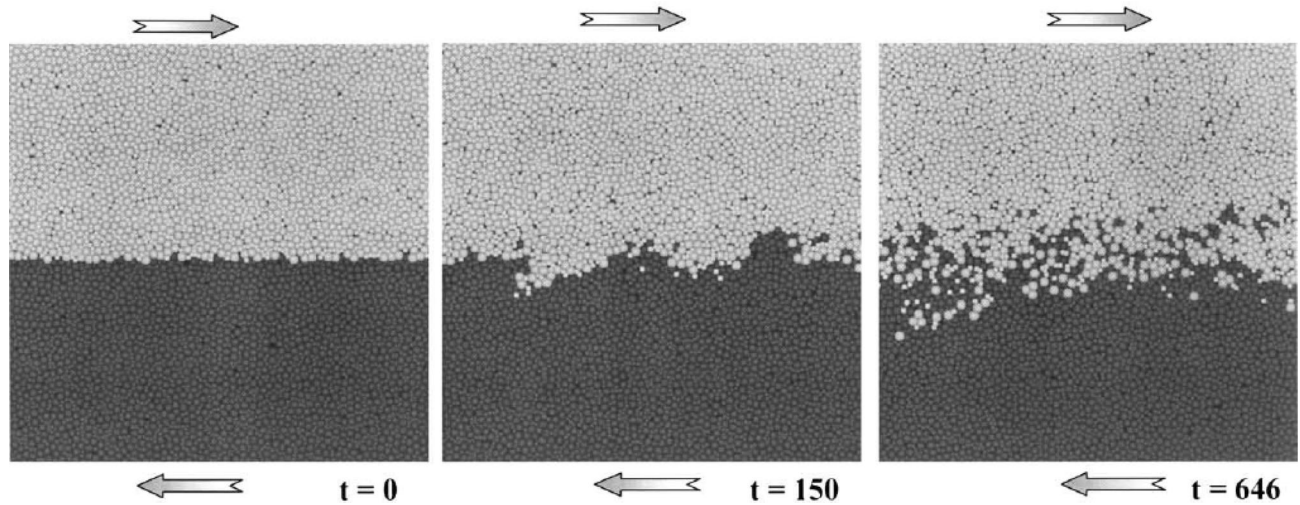


Fig. 4. MD simulations of the evolution of an interface between two glasses subjected to sliding. Reprinted with permission from Ref. 23.

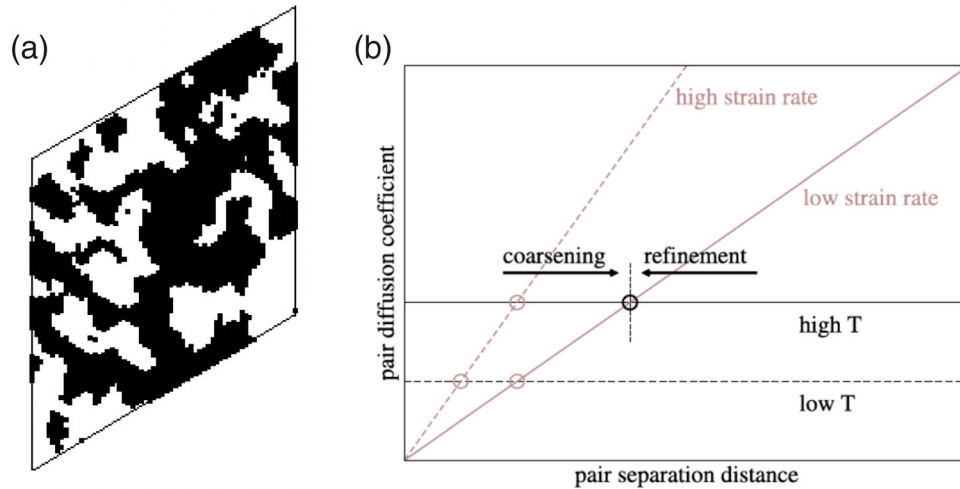


Fig. 5. (a) Monte Carlo simulations of the evolution of an initially perfectly flat (111) interface as in Fig. 1 but now at a finite temperature, resulting in the formation of steady-state self-organized compositional patterns. Reprinted with permission from Ref. 10. (b) Schematic drawing showing the competition between decomposition promoted by thermal diffusion with the atomic mixing forced by plastic deformation. Reprinted with permission from Ref. 24.

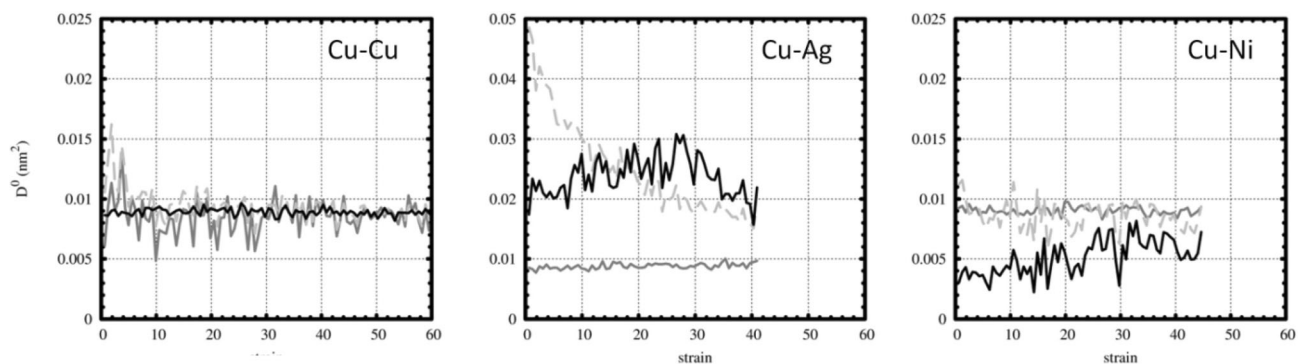


Fig. 6. Effective pair diffusion coefficient of Cu-Cu (solid dark gray), X-X (solid black) and Cu-X (dashed gray) nearest neighbors for X = Cu, Ag, Ni spherical inclusions subjected to bi-axial compression. The effective pair diffusion coefficients are measured per unit strain. Reprinted with permission from Ref. 20.

expected [see Fig. 6a; for the Cu-Ag case, see Fig. 6b], deformation takes place predominantly in the precipitate and at the Cu/Ag interface, whereas for Cu-Ni, deformation is larger in the matrix; see Fig. 6c. These results already indicate that deviations from an ideal geometric mixing are expected for alloys because of differences in their elastic moduli, e.g., deformation tends to localize in the softer phase (the precipitate for Cu-Ag or the matrix for Cu-Ni); see also.²⁵ Differences in the crystallographic structure²⁰ and the alloy heat of mixing^{26,27} will also affect mixing by increasing the stress required to transfer plastic strain across an interface. These effects will be discussed in the following sections.

The simple models and the conclusions provided in this section indicate that it is meaningful to distinguish two regimes when studying the evolution of immiscible alloys subjected to sustained plastic deformation: A low-strain regime, dominated by kinetic roughening and giving rise to large-scale interface features, and a high-strain regime, concerned with chemical mixing at the atomic scale and steady state microstructures. These two regimes are described more completely in Sects. [Low-Strain Regime: Morphological Instabilities](#) and [High-Strain Regime: Forced Atomic Mixing and Self-Organization](#), respectively.

LOW-STRAIN REGIME: MORPHOLOGICAL INSTABILITIES

This section focuses on the low-strain regime, where kinetic roughening is expected to dominate the morphological evolution of two-phase structures. As discussed at the beginning of this Section, strain values in this regime extend to ≈ 10 to 100, although those values will depend significantly upon the initial density of interfaces and the relative out-of-plane to in-plane dislocation activity. These strains are typical of those encountered during many material processing and shaping techniques, e.g., cold rolling, or in some service conditions, for instance materials subjected to sliding wear, as we focus on here. Results reported by Luo et al.²⁸ illustrate very clearly the kinetic roughening of Cu-Au multilayers subjected to reciprocating sliding wear using a nanoindenter; see Fig. 7. This alloy system is particularly advantageous because Cu and Au are both ductile metals, and they have comparable melting points and elastic constants. The chemical ordering tendency in this alloy, moreover, is weak, so that it can be considered a “passive marker” of plastic deformation.²⁹ Figure 7a shows that close to the sliding surface, the initially flat interfaces develop long wavelength undulations at low wear cycles, which evolve into more pronounced roughness at higher cycles as well as vortex-like features; see Fig. 7b as well as Fig. 1 in Ref. 28. Additionally, plastic deformation produces localized thinning and thickening of the layers. We note that

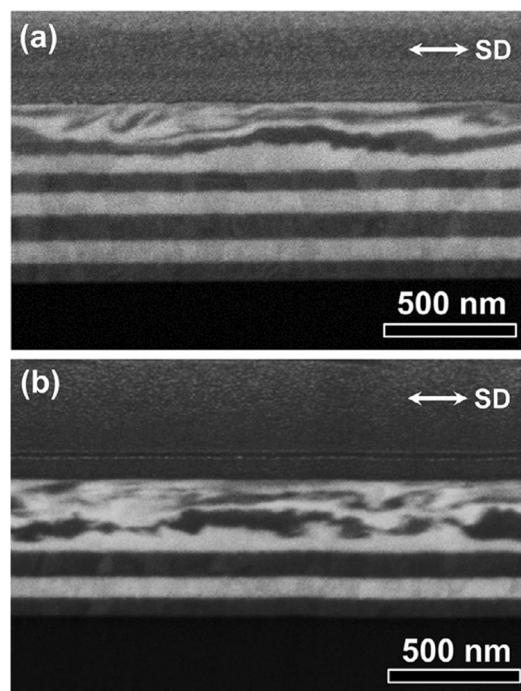


Fig. 7. SEM images of Cu/Au multilayers subjected to reciprocating sliding wear using a conospherical nanoindenter after (a) 50 cycles and (b) 500 cycles. The Au layers (bright contrast) and Cu ones (dark contrast) were initially 102 nm and 110 nm, respectively. Reprinted with permission from Ref. 28.

while similar kinetic roughening is found in the simple geometric model presented in “[Stages of Evolution in Multi-Phase Alloys Subjected to Plastic Deformation](#)” section (Fig. 1), resulting also in localized layer thinning and thickening, no vorticity is detected in the simulations shown in Fig. 1. The lack of vorticity is expected given the imposed homogeneous plastic deformation implemented via a random selection of active slip systems coupled with the constraint of a rigid lattice. Returning to the sliding wear test, another interesting point in those tests is that the plastic strain decreases approximately exponentially with depth, as determined by the displacement of fiducial markers.^{8,30,31} As a consequence, a single cross-section of a worn multi-layered sample, as in Fig. 6, provides a direct visualization of the evolution of interfaces with increasing plastic strain.

The evolution of similar Cu/Au multilayers during sliding wear has been recently studied using MD simulations by Gola et al.³² Due to computational limitations, the individual layer thicknesses were smaller, ≈ 5 nm compared to ≈ 100 nm in the experiments, but the simulations nevertheless generated an evolution that is very similar to the one observed in the experiments (see Fig. 8) with pronounced kinetic roughening after a shear strain of 3. The simulations, in addition, provide unique insights into the conditions required for this evolution to be observed. In particular, while the layers were built close to a (111) orientation, as in Luo’s

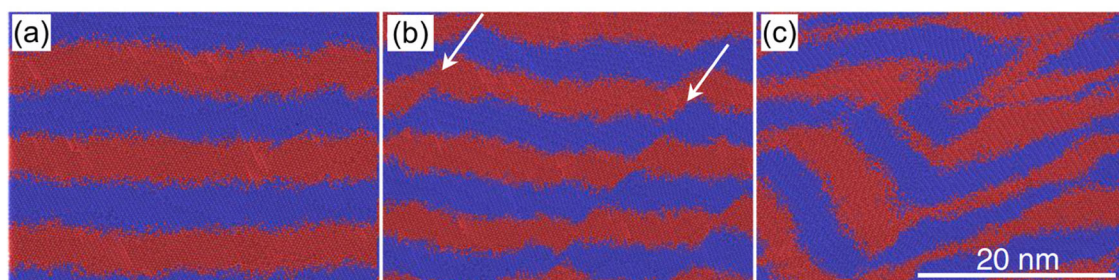


Fig. 8. MD simulations of Cu (blue)/Au (red) multilayers subjected to equi-biaxial stretching to a strain of 0.12 in (a), then 0.25 in (b), followed by simple shear deformation to a strain of 3.0 in (c). Reprinted with permission from Ref. 32 (Color figure online).

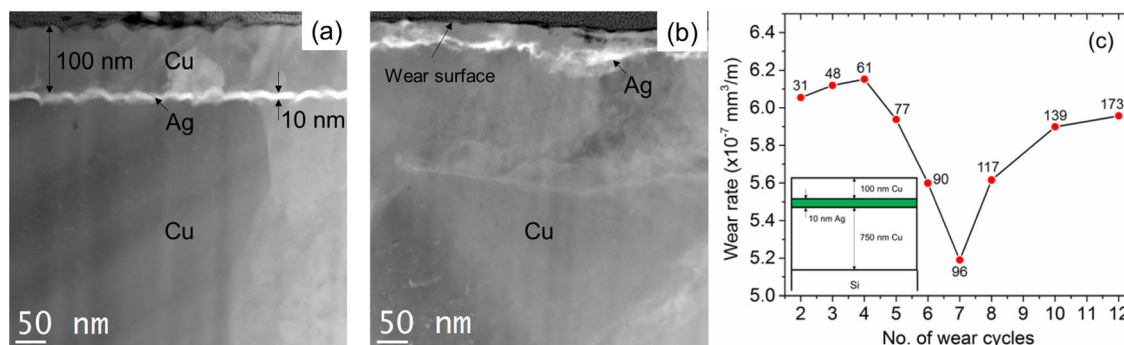


Fig. 9. Z-contrast TEM images from the cross-section of the 100 nm Cu/10 nm Ag/800 nm Cu multilayer after five wear cycles: (a) as-deposited condition and (b) after five wear cycles. (c) Evolution of the wear rate as a function of the number of wear cycle, or as a function of the worn depth as given by the number associated to each data point (red discs). Reprinted with permission from Ref. 33 (Color figure online).

experiments and as commonly observed for fcc thin films grown by PVD, interface misorientation with respect to the exact (111) direction was found to play a critical role (in that work the Cu and Au layers initially share the same crystallographic orientation). If the interface lies exactly in a (111) plane, very limited mixing is observed, and kinetic roughening is absent. This is rationalized by the fact that for an exact (111) interface orientation simple shear can be fully accommodated by in-plane deformation. In the presence of a misorientation, $\tilde{\epsilon}$ in Fig. 8, shear bands form during low equi-biaxial strain; see the arrows in Fig. 8b. Upon subsequent simple shear deformation, these shear bands become the initiation sites of local plastic instability that produces kinetic roughening as well as rotation of portions of grains and thus vorticity.

The evolution of single buried nanolayers during sliding wear offers additional insight into kinetic roughening during wear, as illustrated in the work by Madhavan et al. for the Cu/Ag system.³³ In this study, a three-layer structure with a ≈ 10 nm Ag layer embedded in Cu was grown by PVD; see Fig. 9a. This layer presented some roughness prior to wear testing owing to surface roughening during PVD growth. Figure 9b shows that the Ag layer underwent kinetic roughening and partial dissolution after five wear cycles, i.e., after removal of ≈ 77 nm of the top Cu layer. By comparing single-layer (and multilayer) forced mixing during wear

with forced mixing of a Cu/Ag eutectic by HPT, for which the strain can be measured,¹⁸ it has been estimated that under *steady-state* wear conditions the maximum shear strain at the sliding surface is ~ 350 . The appearance of extensive kinetic roughening and vorticity in Fig. 9b is therefore consistent with the estimates derived in “Stages of Evolution in Multi-Phase Alloys Subjected to Plastic Deformation” section. The single-layer wear test also makes it possible to measure directly the effect of this layer on the wear rate; see Fig. 9c. In this case, Cu/Ag interfaces and solid solution strengthening from Ag mixed into the Cu matrix reduced the wear rate by up to 15%. Notice that the wear rate returned to its reference value only after removal of ≈ 170 nm material, thus far deeper than the initial depth of the buried Ag layer, 110 nm. It was rationalized that shear-induced mixing of Ag into the bottom Cu thick layer and the ensuing solid solution strengthening were responsible for this effect.

The above examples have dealt with two-phase structures where the two phases shared the same fcc lattice structure and were separated by coherent interfaces, allowing dislocations to directly glide across interfaces. Different evolutions are clearly possible when these conditions are no longer met. MD simulations of immiscible A-B binary alloys with bcc precipitates in an fcc matrix²⁰ and of fcc/bcc multilayers^{34–38} illustrate that deformation is often

no longer homogeneous. For highly immiscible systems, A/B interfaces often offer weaker resistance to shearing than the A and B phases, a point that will have important consequences on the large strain response as will be detailed in “[High-Strain Regime: Forced Atomic Mixing and Self-Organization](#)” section. One such alloy system that has received considerable interest is Cu/Nb, in part due to the ability to tailor the mechanical response by controlling the interface crystallography.^{36,39,40} For example, it is observed that for Cu/Nb multilayers grown by PVD the dominant interface is $\{111\}_{\text{Cu}}\langle 110 \rangle_{\text{fcc}}//\{110\}_{\text{Nb}}\langle 111 \rangle_{\text{bcc}}$ with a Kurdjumov-Sachs orientation relationship between the Cu and Nb phases, here referred to as (111)KS, while nanolaminates obtained by accumulative roll bonding (ARB) display interfaces obeying $\{112\}_{\text{Cu}}\langle 111 \rangle_{\text{fcc}}//\{112\}_{\text{Nb}}\langle 110 \rangle_{\text{bcc}}$, also with a KS orientation relationship between the Cu and Nb phases, and thus here referred to as (112)KS. One important difference between the (111)KS and (112)KS interfaces is that the misfit dislocations at these interfaces allow interface slip for (111)KS but not for (112)KS, resulting in large differences in the shear strength of these interfaces, $\approx 200\text{--}400\text{ MPa}$ for (111)KS versus in excess of Pa for (112)KS according to MD simulations.^{35,40} It is therefore interesting to investigate how a (112)KS nanolaminate would respond to shear deformation.

We addressed the above question by subjecting ARB Cu/Nb nanolaminates (with (112)KS interfaces) to sliding wear. Wear testing was performed using a pin-on-disk tester with an alumina disc in ambient air at RT with a load of 9.8 N and a small sliding velocity, 0.05 m/s, to suppress frictional heating.⁴¹ The ARB sample had an individual layer thickness of 20 nm. Steady-state wear rates yielded a low wear coefficient of $1.5 \times 10^{-5} \text{ mm}^3/(\text{N.m})$ compared to a much higher wear coefficient of $1.8 \times 10^{-4} \text{ mm}^3/(\text{N.m})$ measured under the same conditions for a high-performance CuNiSn bronze. Cross sections were prepared for TEM and STEM characterization. The overall morphology of the layers is illustrated in Fig. 10. First, one notices that this macroscopic wear test resulted in plastic deformation up to a depth of $\approx 4\text{ }\mu\text{m}$ below the sliding surface. Upon approaching that surface, the main microstructural modification is the formation of pronounced shear bands at depths ranging from ≈ 4 to $3\text{ }\mu\text{m}$. Closer to the sliding surface, one observes the presence of folds, which we refer to as Z-folds, because of their morphology and their apparent similarity with folds in geological materials.^{13,15} These folds involve many layers, and their typical extension ranges from 0.5 to $1\text{ }\mu\text{m}$. Closer still to the sliding surface, smaller Z-folds or S-shaped swirls are observed, see Fig. 11a, and these structures involve one or just a few individual layers. Lastly, the top $\approx 600\text{ nm}$ of the sample is a nanocomposite with randomly oriented Cu and Nb nanograins, $\approx 10\text{--}20\text{ nm}$ in size; see Fig. 11b. Both types of folds



Fig. 10. A continuous HAADF-STEM image from the depth of 600 nm–4.5 μm below the sliding surface. The arrow indicates the sliding direction. The top 600 nm is comprised of equiaxed nanograins of Cu and Nb. Nb appears brighter than Cu in HAADF mode.

point to a plastic instability in response to the imposed shear deformation. Although the scales are larger than the aforementioned instability reported in the MD simulations for Cu/Au, their morphology suggests that they are also associated with local crystallographic rotations. Electron diffraction patterns (not shown) indicate that new grain orientations are present in the folded regions and that these orientations are distinct from that of the initial Cu and Nb grains. These folds and swirls provide a clue to the strong wear resistance demonstrated by this ARB nanolaminate, as the density of Cu/Nb interfaces near the sliding surface is very high, and their geometry and distribution should suppress shear localization. It is also noteworthy that no crack nucleation was observed in the TEM samples. In agreement with this observation, no large-scale cracks were observed in the Cu/Nb samples at the end of the wear tests, and the wear debris observed by SEM consisted of fine powders, 0.1–1 μm in size. Thus, the wear debris is mainly generated in the topmost region that consists of coexisting Cu and Nb nanograins.

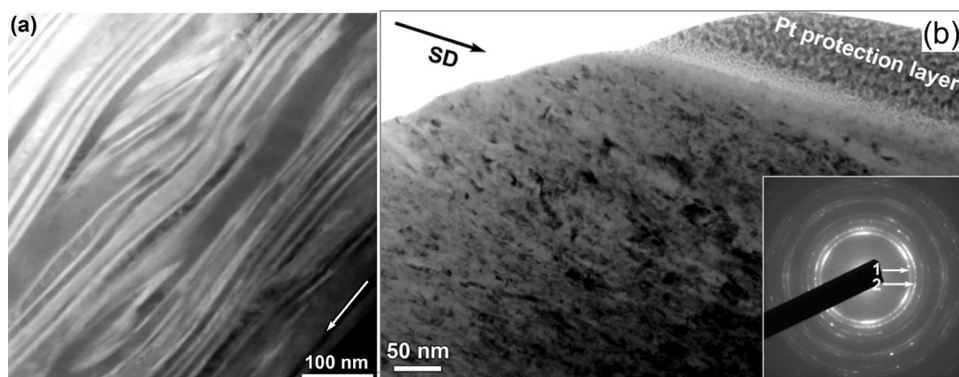


Fig. 11. (a) HAADF-STEM micrographs of swirls seen at a depth of $\sim 1.5 \mu\text{m}$ below the sliding surface. Sliding direction is represented by a white arrow. (b) ND-SD cross-sectional bright-field TEM image and selected area electron diffraction pattern (inset) of the top 600 nm. (1) Nb (110); (2) Cu (111).

The origin of deformation-induced patterns associated with swirls and vorticity has been extensively debated. One possible explanation often proposed, especially in earlier works, is that these patterns form through a Kelvin-Helmholtz (KH) instability, owing to the imposed shear deformation and the different “viscous” deformation rates of the two phases.^{9,42,43} These elements are known to be responsible for swirl patterns in sheared fluids⁴⁴ and in rock formations.^{13,15} More recent studies, for instance Ref. 32, stress however that the shear strains and strain rates required for triggering a KH instability are much larger than the ones reported in experiments. MD simulations³² suggest that these instabilities are intimately related to dislocation interactions with interfaces and other dislocations, and possibly to the contribution of other deformation mechanisms such as twinning. It is also notable that swirl-like patterns have been reported also in other techniques used to impart severe plastic deformation (SPD) such as HPT⁴⁵ and, unsurprisingly, in materials joined by friction stir welding.^{46–48}

We close this section on pattern formation induced by kinetic roughening in layered materials by noting that in a few cases shear deformation leads to just the opposite result, i.e., transforming non-layered structures, e.g., equiaxed precipitates in a matrix, into nanolayered structures,^{49,50} as illustrated in Fig. 12a for Cu-Ag alloys. While the exact origin of this nanolayering reaction is not fully explained, it appears that it requires some contribution of thermal diffusion to increase the stability of interfaces and a favorable shear texture that promotes the localization of the deformation in the sliding direction. KMC simulations using the rigid lattice model described in “Stages of Evolution in Multi-Phase Alloys Subjected to Plastic Deformation” section, but thermal diffusion and preferred in-plane deformation, indeed indicate that wear can induce nanolayering under these particular conditions; see Fig. 12b.

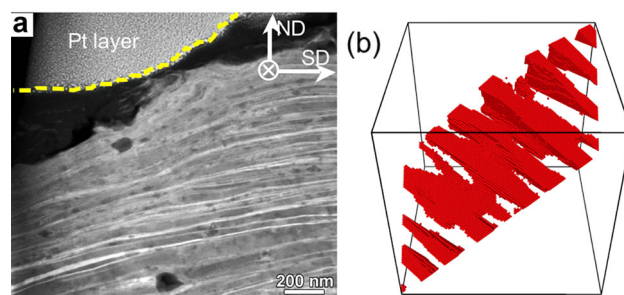


Fig. 12. (a) HAADF-STEM micrograph of sliding wear induced chemical nanolayering in $\text{Cu}_{90}\text{Ag}_{10}$ alloy. Reprinted with permission from Ref. 49. (b) Chemical nanolayering in a model A-B alloy similar to Fig. 1, but with slip localization, with 99.7% shearing in the (111) plane and 0.1% in each remaining {111} slip plane.

HIGH-STRAIN REGIME: FORCED ATOMIC MIXING AND SELF-ORGANIZATION

Effective Temperature Model

The geometric model of shear-induced chemical mixing suggests that alloys homogenize after sufficiently extended plastic deformation at low temperatures. Many studies have demonstrated, however, that alloys with highly reactive or strongly immiscible components acquire a two-phase microstructure during such deformation.^{51–57} Before discussing this work in more detail, we briefly describe Martin’s effective temperature model (ETM) for irradiated alloys,⁵⁸ and a recent modification of it (METM),⁵⁹ to provide a simple explanation for this behavior. The ETM assumes that the phase evolution in driven alloy systems is determined by a competition between thermally activated diffusion processes and forced “ballistic” mixing. The total diffusion flux in a binary A-B alloy is thus given by

$$J_{\text{total}} = J_{\text{th}} + J_b \quad (2a)$$

where

$$J_{th} = -M\nabla\mu = -\frac{c(1-c)\tilde{D}_{th}}{k_B T} \nabla\mu \quad (2b)$$

$$J_b = -D_b \nabla c \quad (2c)$$

where M is the atomic mobility, μ the chemical potential, c the B-species concentration, $\tilde{D}_{th}$ the interdiffusion coefficient and D_b the ballistic diffusion coefficient. Note that the thermal diffusion flux, which takes place at the ambient temperature of the sample, depends on the interdiffusion coefficient, whereas the ballistic mixing obeys Fick's law, as if the sample were at an infinite temperature. The reason for this assumption is that during irradiation, energetic recoils are several tens of eVs or more, so that chemical interactions can be neglected. Martin then assumed that $J_{total} = 0$ when the system was in steady state, which enabled him to define an *effective potential* for the irradiated system that plays the same role as the free energy for an equilibrium system.⁵⁸ For a regular solution model for the free energy, this leads to the rule of corresponding states, i.e., a system driven at temperature, T , behaves the same as if the system were at equilibrium but at some higher temperature, T_{eff} , where $T_{eff} = T(1 + \gamma_b)$, where $\gamma_b = D_b/\tilde{D}_{th}$. Atomic mixing during irradiations, however, occurs largely in thermal spikes rather than by energetic recoil events, and so forced atomic mixing can be affected by chemical interaction in strongly immiscible and reactive systems. During severe plastic deformation, atoms are similarly forced over potential barriers at relatively low energies,⁶⁰ and so chemical biasing can again be important. This can be realized by noting that the work done per atom during shear deformation, w , is $w = \sigma\varepsilon V_0$, where σ is the applied stress, ε is the strain, and V_0 is the atomic volume. By assuming the maximum average stress is given by the theoretical strength, ~ 10 GPa, and noting that each atom makes $\sim$ one jump after a strain of unity, we estimate that the work done per atom, per jump, is ~ 0.5 eV. This energy is typical of chemical interactions for strongly immiscible or highly reactive alloy components.²⁷ To account for these interactions, Verma et al. extended the Martin model by replacing the purely ballistic solute flux, Eq. 2c, with $J_b^0 = -M_b^0 \nabla\mu(T_0) = -\frac{c(1-c)D_b^{T_0}}{k_B T} \nabla\mu(T_0)$, where M_b^0 and $D_b^{T_0}$ are the mobility and the diffusion

coefficient attributed to the external driving forces, and T_0 is the characteristic temperature of the mixing process.⁵⁹ As an example, T_0 for irradiation would be roughly the thermal spike temperature, while for plastic deformation it would be on the order of w/k_B . With this modification, a new effective potential can be determined, and in the regular solution model,

$$T_{eff} = T \left(1 + \gamma_b^{T_0} \right) / \left(1 + \frac{T}{T_0} \gamma_b^{T_0} \right) \quad (3)$$

where $\gamma_b^{T_0} = D_b^{T_0}/\tilde{D}_{th}$. Now, as $T \rightarrow 0$, $T_{eff} \rightarrow T_0$, i.e., within the METM model, the phases present in the alloy after extended plastic deformation at low temperature will be given by the equilibrium phase diagram at temperature T_0 . This is significant since plastic deformation at room temperature approximates the low temperature limit as vacancies in many alloys are immobile at 300 K, and hence $\tilde{D}_{th} = 0$.

The METM has been examined by both experiments and by MD simulation. Beach et al.⁶¹ for example, measured the concentration of Nb in solution in Cu during low-temperature (-78°C) deformation using high-pressure torsion (HPT). Initially, the alloys were either solid solutions (grown by magnetron sputtering) or two phase (pre-annealed films). The results are reported in Fig. 13 where it is observed that Nb precipitates out of solution if the initial Nb concentration is

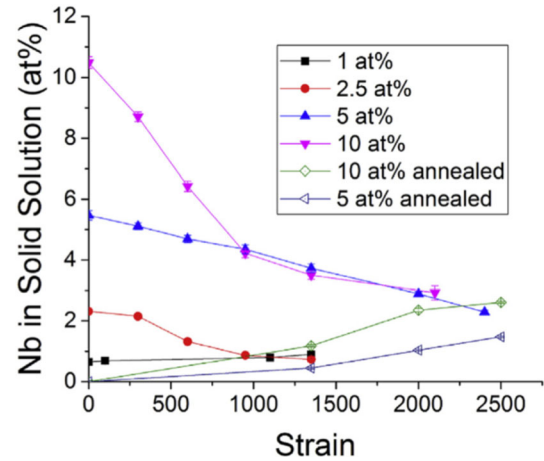


Fig. 13. Solubility of Nb in Cu versus strain. Solid symbols: HPT of as-deposited samples; open circles: HPT of pre-annealed samples. Reprinted with permission from Ref. 61.

Table I. Calculated solubilities of various alloying components in Cu assuming $T_0 = 3000$ K. Adapted from Ref. 59

	Mo	Nb	V	Ta	Cr	W	Co	Ag	Ni
$\alpha(kJ/mol)$	65	63	117		42	100	30	15	10
$\alpha + \Delta G_X^{FCC}(kJ/mol)$	106	96	141	164	80	146	30	15	10
X_{sat}	0.014	0.021	0.0034	0.0014	0.04	0.003	SS	SS	SS

≥ 2 at.% and Nb precipitates will dissolve if the initial concentration is $\lesssim 2$ at.%. Thus, the system has a unique steady-state solubility. For these dilute solutions, the equilibrium solubility is given by $X_{sol} = \exp(-\Delta H_s/k_B T)$, where ΔH_s is the heat of solution. For Nb in Cu, $\Delta H_s \sim 96 \text{ kJ/mol}$, and so the experimental value $X_{sol} = 0.020$ implies $T_0 \sim 3,100$ K. Similar experiments performed on dilute Cu-Mo alloys showed quite similar results, $T_0 \sim 2650$,⁵⁹ the difference between these two values of T_0 falls within the uncertainties in ΔH_s . With the assumption that $T_0 = 3000$ K for Cu alloys, Verma calculated the solubilities of a number of immiscible Cu alloys as shown in Table I using a regular solution model and compared the results with literature data. Noteworthy is that Co, Ag and Ni, for which $\Delta H_s < 30 \text{ kJ/mol}$, all form complete solute solutions with Cu while the other alloys, for which $\Delta H_s > 80 \text{ kJ/mol}$, all form two-phase alloys. These predictions are confirmed experimentally.⁵⁹

Atomic Processes: MD Simulation

While the METM provides a firm basis for rationalizing the general trends of phase evolution during severe plastic deformation, MD offers an atomistic description of SPD that helps in understanding both the detailed processes of forced chemical mixing in strongly immiscible alloys and the limitations of the METM. Unlike experiments on real systems, interatomic potentials can be created that isolate different properties of alloys. We illustrate this by describing MD studies on SPD in two strongly immiscible alloys. The first examines the Cu(1)-Cu(2). In this *pseudo alloy* the interactions between Cu(1) atoms and between Cu(2) atoms are identical to those in “real” Cu as described by the interatomic potential of Cleri-

Rosato.⁶² The interactions between Cu(1) and Cu(2) atoms in this model, on the other hand, were modified to create a large heat of solution, 2 eV, at infinite dilution, but with little change in the elastic properties.²⁷ Plastic deformation was effected in these simulations by applying biaxial compressions, using a three-step cycle to maintain the original cubic shape of the computational cell; each cycle imparts a strain of 1.2.

The results are shown in Fig. 14a for an equiatomic Cu(1)-Cu(2) alloy (bottom curve) where the average alloy concentration in solution is plotted as a function of strain. This simulation was initiated as a bilayer sample, but similar to the experiments, the steady-state solubility is independent of whether the initial state of the alloy is a bi-layer

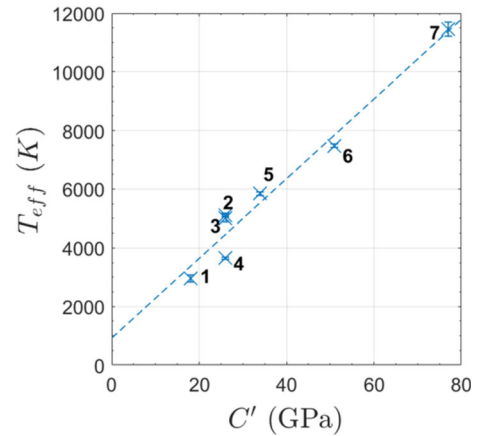


Fig. 15. Plot of the effective temperature, T_{eff} , versus tetragonal shear modulus, C' , for X(1)-X(2) alloys. Numbers identify the pseudo alloy as X-X/ Ω , where Ω is the heat of mixing of the X(2)-X(1) alloy at infinite dilution. (1: Ag/1.6 eV; 2: Pd/1.5 eV; 3: Cu/1.7 eV; 4: Cu/2.0 eV; 5: Pt/2.0 eV; 6: Ni/2.0 eV; 7: Rh/3.2 eV. Adapted from Ref. 27.

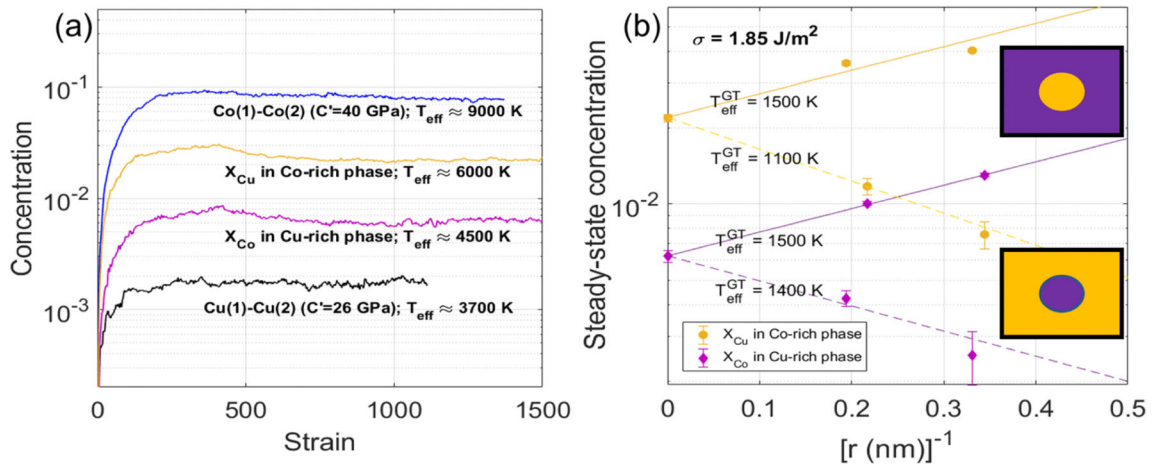


Fig. 14. (a) Solute concentrations in Cu(1)-Cu(2), Co(1)-Co(2) and Cu-Co pseudo-alloys plotted as a function of strain. All alloys have a heat of mixing $\Omega = 2 \text{ eV/atom}$ and a bilayer configuration. Indicated values of T_{eff} refer to a regular solution model. (b) Steady-state solute concentrations in the Cu-Co ($\Omega = 2 \text{ eV/atom}$) system containing a single precipitate, plotted as a function of inverse precipitate radius. The solubilities at $r^{-1} = 0$ correspond to the data for a planar interface. Indicated values of T_{eff}^{GT} were calculated using a value of $\sigma = 1.85 \text{ J/m}^2$ for the interface free energy. Adapted from Ref. 63.

or a random solid solution. Moreover, the steady-state solubility, 0.19 at.%, yields an effective temperature of $T_0 \sim 3700$ K, which is remarkably close to that found in the experiments. The simulations were repeated for a number of other fcc metals, with the results for T_0 plotted in Fig. 15 as a function of the tetragonal shear modulus. The correlation between T_0 and C' is observed to be quite good, providing support for our physical interpretation of T_0 in the METM, i.e., that T_0 should scale with the shear modulus. The MD simulations also reveal some difficulties with the METM. Simulations of SPD in Cu(1)-Cu(2) with heats of solution, ΔH_s , < 1.4 eV, for example, do not lead to phase separation but rather to short range ordering. While this result is not unexpected, the value of T_0 deduced from these simulations increases to ~ 7000 K, which is significantly higher than that deduced from the solubility. The jump in T_0 from 3700 K to 7000 K occurs at values of ΔH_s just where the alloy makes the transition from two-phase to a solid solution, indicating that interfaces play an important role in determining T_0 . Since interfaces in these strongly immiscible alloys are weaker than the perfect crystal, the local stresses at the interphase are likely reduced, and so it is reasonable that T_0 should be reduced. It shows, however, that multiphase systems must be described by different local effective temperatures, which can greatly complicate the application of the METM.

We next consider the mixed pseudo alloy, Cu-Co, to examine the effect of different elastic properties in the two terminal phases on phase evolution. We

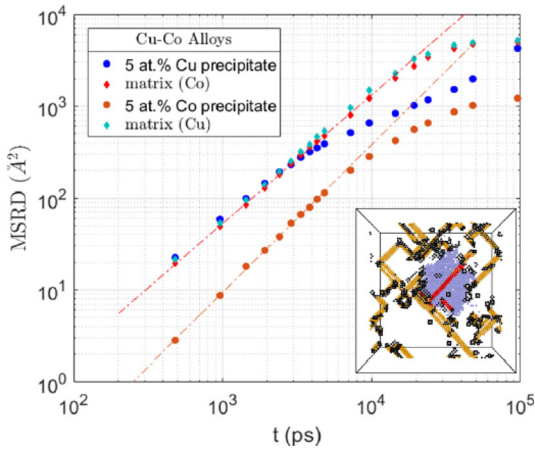


Fig. 16. MSRD displacements for either a 5 at.% Cu embedded in a Co matrix, or vice versa, with heat of mixing of $\Omega = 2 \text{ eV/atom}$ for all systems. The Cu particle is seen to co-deform with the Co matrix, while there is significant shear localization in the Cu matrix in the Cu-0.05 Co alloy. Inset shows a cross-section of the simulation cell featuring a Co precipitate in the Cu matrix with the coloration indicating the local atomic environment. The Cu fcc atoms have been omitted for clarity, while the Co fcc atoms are colored blue. Atoms with HCP symmetry are colored yellow or red for the Cu and Co species, respectively; they indicate the glide of a partial dislocation on that plane. Adapted from Ref. 63.

begin with a bilayer sample as the initial state, with $\Delta H_s = 2$ eV, and plot as a function of strain in Fig. 14a the concentration of Cu and Co in the Co-rich and Cu-rich fcc phases, respectively. Also shown for comparison are corresponding results for Cu(1)-Cu(2) and Co(1)-Co(2). Notice that the solubility of Cu in Co is less than is Co(2) in Co(1) even though ΔH_s is the same for the two systems. These results would indicate in the METM that T_0 is greater in the Co rich phase than in the Cu rich phase, as expected from Fig. 15, but the presence of the Cu-rich phase lowers T_0 in the Co-rich phase, whereas the Co-rich phase increases T_0 in the Cu-rich phase. The mixed Cu-Co system was also initiated with a spherical Co (or Cu) precipitate of different sizes within a Cu (or Co) matrix. Results for the alloy concentrations in these cases are shown in Fig. 14b as a function of the inverse radius of the precipitate. Remarkably, the two systems display behavior reminiscent of the Gibbs-Thomson effect in systems at equilibrium, but see Ref. 63 for details. Similar to Fig. 6, we plot in Fig. 16 the relative pair diffusion coefficients for nearest neighbor atoms in the two matrices and in the precipitates as a function of strain. The pair diffusion coefficients in the Cu and Co matrices are the same, and nearly so for the pair diffusion coefficient in the Cu precipitate, but the pair diffusion coefficient in the Co precipitate is reduced by a factor of ~ 5 . This latter result can be understood from the fact that the Co particle is harder than the Cu matrix, so that strain is localized in the Cu matrix. The more interesting point, however, is that the reduced strain in the Co precipitate has no apparent effect on the solubility of Cu in the Co precipitate; see Fig. 14. This suggests that the steady-state solubility is not

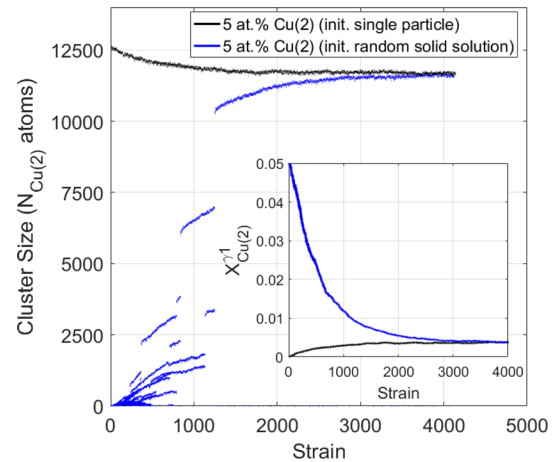


Fig. 17. Sizes of individual clusters plotted as a function of strain for a Cu(1)-5 at.% Cu(2) alloy. Initial conditions are a random alloy (blue points) or phase-separated alloy (black points). Inset shows the concentration of Cu(2) in solution as a function of strain for the two different starting configurations. Adapted from Ref. 63 (Color figure online).

dependent on local strain, or $D_b^{T_0}$, but only on the effective temperature of the phase.

While the steady-state phases present during low temperature SPD are not directly related to $D_b^{T_0}$, the evolution of these phases illustrates the superdiffusive character of $D_b^{T_0}$ discussed in “[Stages of Evolution in Multi-Phase Alloys Subjected to Plastic Deformation](#)” section. As noted there, the relative motion of two particles increases proportionally to their distance apart; this is true for pairs of clusters as well as for pairs of atoms. Consequently, precipitate growth is strongly affected by cluster agglomeration. This can be observed in MD simulations of nucleation and growth of Cu(2) precipitates during shearing of a Cu(1)-5 at.% Cu(2) solid solution. As seen in Fig. 17, many precipitates nucleate and grow, but as they grow larger, agglomeration becomes the predominant mechanism for growth. After strain of ~ 1300 in this simulation, only a single precipitate remains in the computational cell and no further growth is possible. Also shown in Fig. 17 is the Cu(2) concentration dissolved in Cu(1), $X_{Cu(2)}$, plotted as a function of strain. It is observed that $X_{Cu(2)}$ saturates at a strain of ~ 3000 , which is just the strain where the Cu(2) precipitates stops growing. Comparing experiments with these simulations, it is seen in Fig. 13 that the concentration of Nb dissolved in Cu during HPT also stabilizes after a strain of 3000. This good agreement indicates that the microstructural evolution is not overly sensitive to the details of the strain path or strain rate.

While the MD simulations show that precipitates continue to coarsen until just a single precipitate remains, the experiments reveal that precipitates adopt a steady-state size that is a characteristic of the alloy system. For Cu-5 at.%Nb and Cu- 5 at.% Mo, this size is ~ 15 nm in diameter,^{59,61} i.e., if the initial state has larger precipitates, they shrink, while smaller precipitates grow. Since 15 nm size is larger than the size of the MD computational box size, there is no contradiction between the experimental and MD results, but nevertheless, precipitate size selection, i.e., self-organization, is not predicted by the METM. Currently, there is no accepted explanation for self-organization during SPD at low T, but we draw attention to the unique feature of superdiffusive mass transport during plastic deformation to provide some rationalization for this behavior. Consider first a supersaturated solution subjected to SPD. The MD simulations (Fig. 17) show that precipitates nucleate and grow by a combination of agglomeration and “evaporation-condensation” mechanisms, but that agglomeration dominates at larger sizes. This efficient growth mechanism, as noted above, is limited by the grain size, and thus the grain size sets an upper limit to the growth of the precipitates. The experiments on Cu-Nb and Cu-Mo, for example, show that in steady state the precipitates are $\sim 1/3$ of the grain size. Next consider the shrinkage of

very large precipitates some microns in size. These cannot shrink by simple erosion as shown in Fig. 17 for a small precipitate; this mechanism is far too slow. As discussed in “[Low-Strain Regime: Morphological Instabilities](#)” section, however, kinetic roughening initially controls phase morphology, and as shown in Figs. 7, 8 and 11b, even laminar precipitates can be broken up into smaller spherical units with sizes comparable to the grain size. These much smaller precipitates can then be further reduced by erosion. The diffusion model assumed in the METM does not include kinetic roughening, and therefore it cannot explain this effect.

CONCLUSION

In this work, we proposed to separate the evolution of microstructures in multi-phase alloys into two regimes, a low-strain regime dominated by shear-induced superdiffusive mixing and resulting in kinetic roughening of interfaces, and fragmentation of precipitate phases, and a large-strain regime dominated by force atomic mixing and responsible for nonequilibrium solubilities. We illustrated that the former regime is particularly relevant to materials undergoing wear since continuous removal of material at the wear surface prevents the plastic strain from reaching very large values. In addition to the kinetic roughening predicted by simple geometric models, chemical and structural effects can produce complex interface instabilities, e.g., grain rotation, folds and swirls that, in some alloys, result in improved wear resistance. In the large strain regime, we showed that a modified effective temperature criterion makes it possible to rationalize the effect of immiscibility and elastic contrast on nonequilibrium solubility. Furthermore, this criterion is given a firm physical footing by introducing the concept of a process temperature for the forced mixing that is related to the plastic work done per atom during shearing. Lastly, combining these two main effects, superdiffusive mixing and forced atomic mixing, is shown to provide a convincing rationalization for the stabilization of steady-state compositional patterns in nanocrystalline materials. It is suggested that, for a given shaping process or a given application, the understanding afforded by the present two-regime approach helps identify the most important effects and that it should therefore facilitate improving existing materials or designing new high-performance materials for specific conditions.

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CONFLICT OF INTEREST

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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